

SAINT-LOUIS, Senegal

WMO: 616000

Lat: 16.051N

Lon: 16.463W

Elev: 3

StdP: 101.29

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	15.9	-1.2	3.4	27.0	0.8	4.0	25.4	8.4	24.9	7.4	25.4	2.4	40	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	6.6	38.8	21.6	36.9	21.2	35.1	21.3	28.5	31.9	28.1	31.2	27.7	30.7	4.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.8	23.8	30.4	27.2	23.0	29.8	26.9	22.6	29.6	92.1	32.1	90.0	31.4	88.0	31.2	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.8	6.2	DB	12.1	42.8	1.5	1.1	11.0	43.6	10.2	44.2	9.4	44.8	8.3	45.6	
			WB	8.0	29.9	1.0	1.2	7.2	30.8	6.6	31.5	6.0	32.2	5.3	33.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.5	23.9	24.4	25.0	24.6	24.8	26.6	28.0	28.7	29.1	29.5	27.7	25.0
	DBStd	2.89	2.28	2.55	3.12	2.77	2.38	1.73	1.16	1.05	1.28	1.88	1.94	2.18
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6005	429	404	464	438	459	498	559	579	574	604	531	464
	CDD18.3	2964	172	171	206	188	201	248	301	321	324	346	281	206
	CDH23.3	28347	1412	1444	1679	1291	1325	2083	3216	3800	3919	3952	2554	1672
	CDH26.7	11850	641	698	851	552	445	611	1039	1467	1665	1809	1294	778
(14) Wind	WSAvg	3.5	3.3	3.8	4.1	4.5	4.2	3.7	3.5	3.1	2.7	3.0	3.0	3.1
(15) Precipitation	PrecAvg	251	1	1	0	0	0	7	45	93	104	18	1	1
	PrecMax	532	21	14	5	3	3	70	207	242	380	198	15	33
	PrecMin	58	0	0	0	0	0	0	0	12	4	0	0	0
	PrecStd	98	4	3	1	0	0	12	42	51	73	35	3	5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	37.9	39.9	41.3	38.9	35.5	33.9	34.4	35.8	41.0	39.8	37.1
		MCWB	19.6	19.4	20.2	21.9	21.3	24.5	26.5	27.1	27.3	23.6	21.6	19.9
	2%	DB	34.0	35.5	37.2	35.1	33.3	32.3	32.2	32.7	33.8	37.6	37.3	34.9
		MCWB	18.5	18.8	20.1	21.0	22.1	25.0	26.5	27.3	27.6	22.9	21.1	19.7
	5%	DB	32.3	33.5	34.6	31.9	30.8	30.8	31.2	32.0	32.9	34.8	35.4	33.1
		MCWB	18.0	18.1	19.1	21.0	22.3	25.1	26.3	27.0	27.6	25.1	20.8	19.2
	10%	DB	30.6	31.2	31.5	29.4	28.8	29.8	30.7	31.2	32.0	33.1	33.5	31.1
		MCWB	17.6	17.7	19.1	20.9	22.3	24.8	26.1	26.8	27.3	26.3	21.3	18.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	22.5	23.9	24.2	25.4	26.8	28.1	28.6	29.2	29.2	26.9	24.7
		MCDB	31.1	30.5	32.9	34.3	30.2	30.7	31.4	32.1	32.6	32.8	31.1	31.0
	2%	WB	21.0	21.3	22.4	22.9	24.4	26.2	27.2	28.0	28.6	28.4	25.8	23.2
		MCDB	29.0	29.4	31.4	30.4	28.9	29.8	30.5	31.1	31.9	31.9	29.9	28.5
	5%	WB	20.1	20.5	21.4	22.1	23.6	25.7	26.8	27.6	28.2	27.9	24.7	22.2
		MCDB	27.8	28.4	29.2	29.0	27.9	29.1	29.9	30.6	31.2	31.1	29.3	27.5
	10%	WB	19.2	19.6	20.5	21.3	22.9	25.3	26.5	27.2	27.8	27.3	23.8	21.1
		MCDB	26.2	27.0	27.5	27.5	27.0	28.5	29.5	30.1	30.6	30.3	28.4	26.8
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	13.7	12.9	11.1	8.9	6.7	5.6	5.9	6.6	9.0	12.8	13.6
		MCDBR	16.3	16.9	17.8	15.2	12.7	8.7	6.6	6.8	7.7	12.5	15.8	16.1
		MCWBR	7.3	6.6	5.9	5.0	4.5	3.1	2.7	2.9	3.3	4.3	5.9	6.8
	5% WB	MCDBR	14.1	13.7	13.7	12.3	9.3	6.9	6.1	6.4	6.8	8.4	12.3	13.4
		MCWBR	6.9	6.1	5.6	4.9	4.1	2.9	2.8	3.2	3.4	3.8	5.4	6.5
(40) Clear-Sky Solar Irradiance	taub		0.541	0.556	0.612	0.636	0.703	0.768	0.746	0.675	0.640	0.614	0.518	0.519
	taud		1.808	1.786	1.686	1.693	1.570	1.530	1.594	1.742	1.801	1.814	1.964	1.913
	Ebn at Noon		745	760	732	716	660	611	627	679	702	705	762	751
	Edh at Noon		206	220	250	249	277	286	268	234	220	211	175	181
(44) All-Sky Solar Radiation	RadAvg		4.99	5.63	6.60	7.12	7.08	6.62	6.18	5.97	5.76	5.76	5.13	4.74
	RadStd		0.35	0.32	0.23	0.24	0.27	0.16	0.19	0.19	0.24	0.24	0.28	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page